

LAB02

Tarefa

```
library(readr)
library(dplyr)
library(ggplot2)
library(ggcal)
```

```
processa_chunk <- function(df, pos) {
  df %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) %>%
    mutate(
      DATE = as.Date(paste(YEAR, MONTH, DAY, sep = "-")),
      atrasado = ifelse(ARRIVAL_DELAY > 10, 1, 0)
    ) %>%
    filter(!is.na(ARRIVAL_DELAY)) %>%
    group_by(DATE, AIRLINE) %>%
    summarise(
      voos_atrasados = sum(atrasado),
      voos_totais = n(),
      .groups = "drop"
    )
}
resultados_parciais <- read_csv_chunked(
  file = "flights.csv",
  callback = DataFrameCallback$new(processa_chunk),
  chunk_size = 100000
)
```

```
-- Column specification -----
cols(
  .default = col_double(),
  AIRLINE = col_character(),
  TAIL_NUMBER = col_character(),
  ORIGIN_AIRPORT = col_character(),
  DESTINATION_AIRPORT = col_character(),
  SCHEDULED_DEPARTURE = col_character(),
  DEPARTURE_TIME = col_character(),
  WHEELS_OFF = col_character(),
  WHEELS_ON = col_character(),
  SCHEDULED_ARRIVAL = col_character(),
  ARRIVAL_TIME = col_character(),
  CANCELLATION_REASON = col_character()
)
i Use `spec()` for the full column specifications.
```

```
resultados_finais <- resultados_parciais %>%
  group_by(DATE, AIRLINE) %>%
  summarise(
    total_atrasos = sum(voos_atrasados),
    total_voos = sum(voos_totais),
    .groups = "drop"
  ) %>%
  mutate(percentual_atraso = (total_atrasos / total_voos) * 100)

head(resultados_finais)
```

```
# A tibble: 6 x 5
  DATE      AIRLINE total_atrasos total_voos percentual_atraso
  <date>    <chr>      <dbl>      <int>      <dbl>
1 2015-01-01 AA          514        1379        37.3
2 2015-01-01 DL          132        1558         8.47
3 2015-01-01 UA          367        1269        28.9
4 2015-01-01 US          201        1028        19.6
5 2015-01-02 AA          649        1508        43.0
6 2015-01-02 DL          406        2261        18.0
```

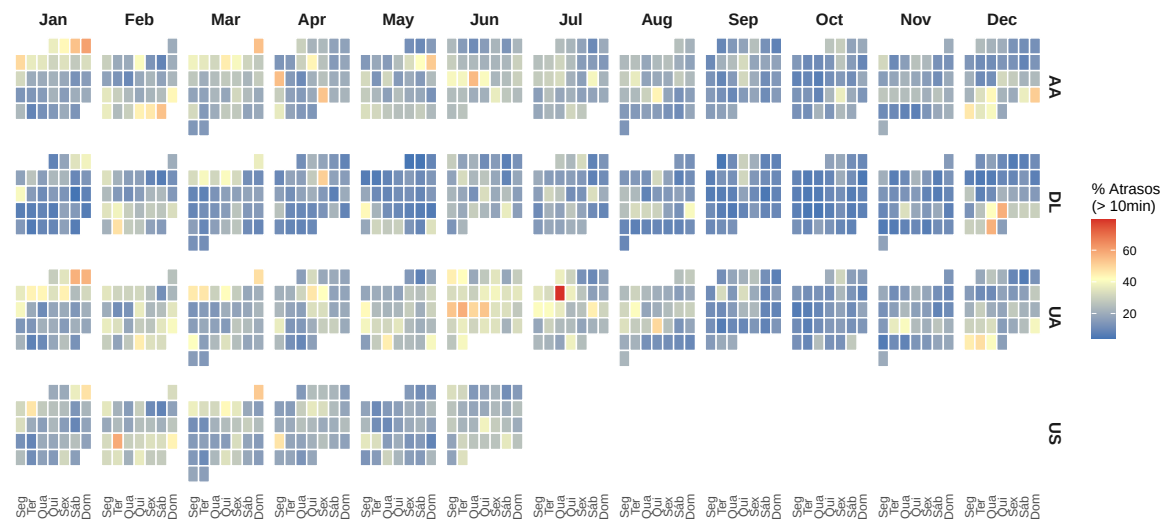
```
dados_calendario <- resultados_finais %>%
  filter(!is.na(DATE)) %>%
  mutate(
    mes = as.integer(format(DATE, "%m")),
```

```

mes_nome = factor(mes, levels = 1:12, labels = month.abb),
dia_semana = factor(format(DATE, "%u"), levels = 1:7,
                      labels = c("Seg", "Ter", "Qua", "Qui", "Sex", "Sáb", "Dom")),
semana_ano = as.numeric(format(DATE, "%W"))
) %>%
group_by(mes_nome) %>%
mutate(semana_mes = semana_ano - min(semana_ano, na.rm = TRUE) + 1) %>%
ungroup()
ggplot(dados_calendario, aes(x = dia_semana, y = semana_mes, fill = percentual_atraso)) +
  geom_tile(color = "white", linewidth = 0.5) +
  facet_grid(AIRLINE ~ mes_nome) +
  scale_y_reverse() +
  scale_fill_gradientn(colors = c("#4575b4", "#ffffbf", "#d73027"),
                       name = "% Atrasos\n(> 10min)") +
  labs(
    title = "Percentual de Voos com Atraso na Chegada (> 10 min) - Ano 2015",
    subtitle = "Companhias: AA, DL, UA e US",
    x = "",
    y = ""
  ) +
  theme_minimal() +
  theme(
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    panel.grid = element_blank(),
    strip.text.x = element_text(face = "bold", size = 12),
    strip.text.y = element_text(face = "bold", size = 12),
    axis.text.x = element_text(angle = 90, vjust = 0.5)
  )

```

Percentual de Voos com Atraso na Chegada (> 10 min) – Ano 2015
 Companhias: AA, DL, UA e US



1)

Precisamos de apenas dois valores: a contagem de voos com atraso superior a 10 minutos e a contagem total de voos avaliados.

2)

```
getStats <- function(input, pos) {
  input %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) %>%

    filter(!is.na(ARRIVAL_DELAY)) %>%

    group_by(DAY, MONTH, AIRLINE) %>%

    summarise(
      voos_atrasados = sum(ARRIVAL_DELAY > 10),
      voos_totais = n(),
      .groups = "drop"
    )
}
```

3)

```
colunas_selecionadas <- cols_only(  
  MONTH = col_integer(),  
  DAY = col_integer(),  
  AIRLINE = col_character(),  
  ARRIVAL_DELAY = col_double()  
)  
resultados_agrupados <- read_csv_chunked(  
  file = "flights.csv.zip",  
  callback = DataFrameCallback$new(getStats),  
  chunk_size = 100000,  
  col_types = colunas_selecionadas  
)  
head(resultados_agrupados)
```

```
# A tibble: 6 x 5  
  DAY MONTH AIRLINE voos_atrasados voos_totais  
  <int> <int> <chr>          <int>      <int>  
1     1     1 AA             514      1379  
2     1     1 DL             132      1558  
3     1     1 UA             367      1269  
4     1     1 US             201      1028  
5     2     1 AA             649      1508  
6     2     1 DL             406      2261
```

4)

```
computeStats <- function(df) {  
  df %>%  
    group_by(MONTH, DAY, AIRLINE) %>%  
    summarise(  
      soma_atrasados = sum(voos_atrasados),  
      soma_totais = sum(voos_totais),  
      .groups = "drop"  
    ) %>%  
    mutate(  
      Perc = soma_atrasados / soma_totais,
```

```

    Data = as.Date(paste("2015", MONTH, DAY, sep = "-")),
    Cia = AIRLINE
  ) %>%
  select(Cia, Data, Perc)
}

```

5)

```

pal <- scale_fill_gradient(low = "#4575b4", high = "#d73027", name = "% Atraso")

baseCalendario <- function(stats, cia) {
  df_cia <- filter(stats, Cia == cia)
  base <- ggcal(df_cia$Data, df_cia$Perc)
  return(base)
}

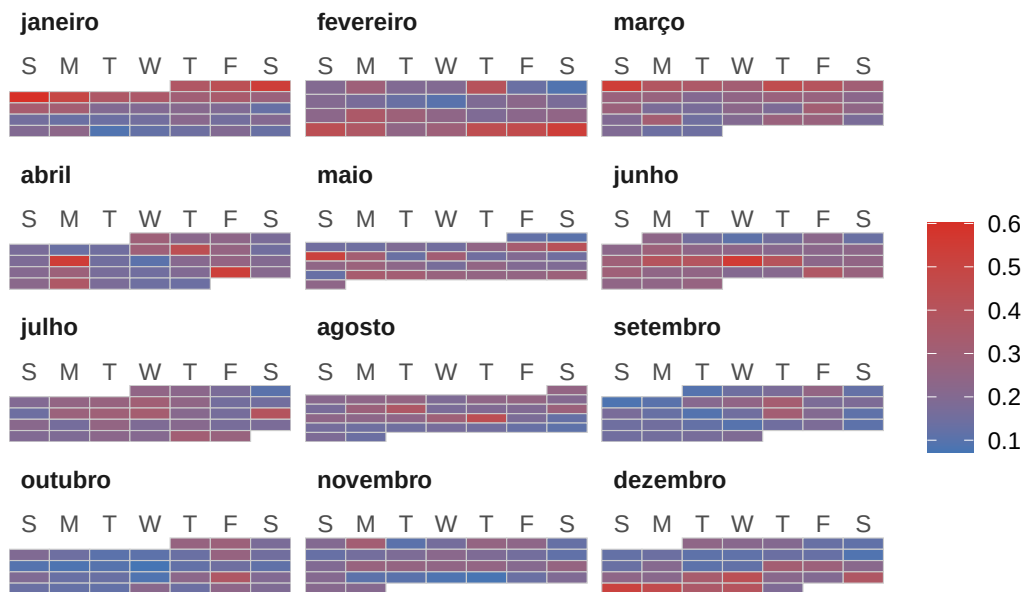
dados_para_calendario <- computeStats(resultados_agrupados)

cAA <- baseCalendario(dados_para_calendario, "AA")
cDL <- baseCalendario(dados_para_calendario, "DL")
cUA <- baseCalendario(dados_para_calendario, "UA")
cUS <- baseCalendario(dados_para_calendario, "US")

cAA + pal + ggtitle("Calendário de Atrasos - American Airlines (AA)")

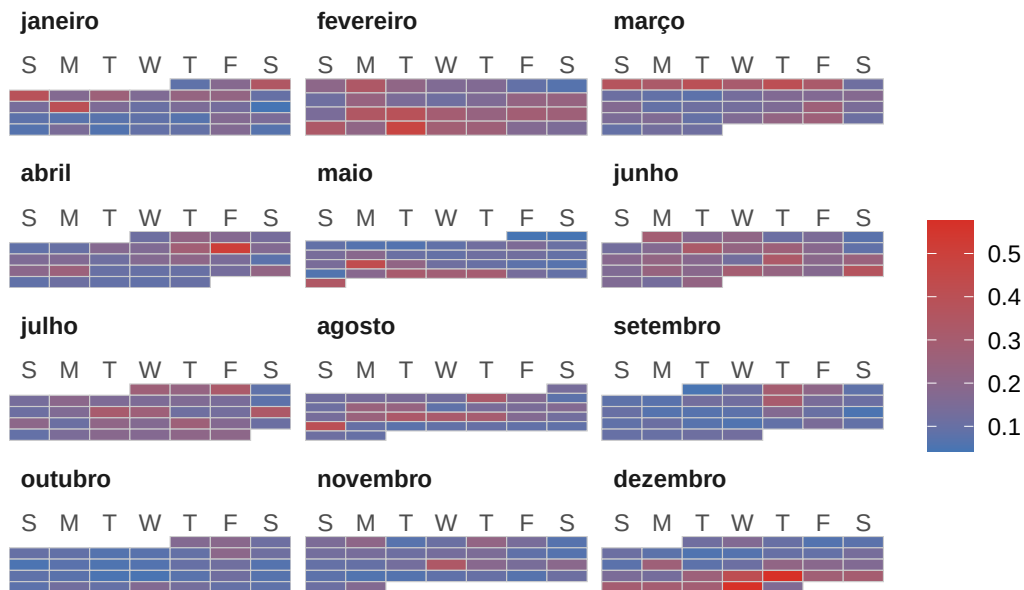
```

Calendário de Atrasos – American Airlines (AA)



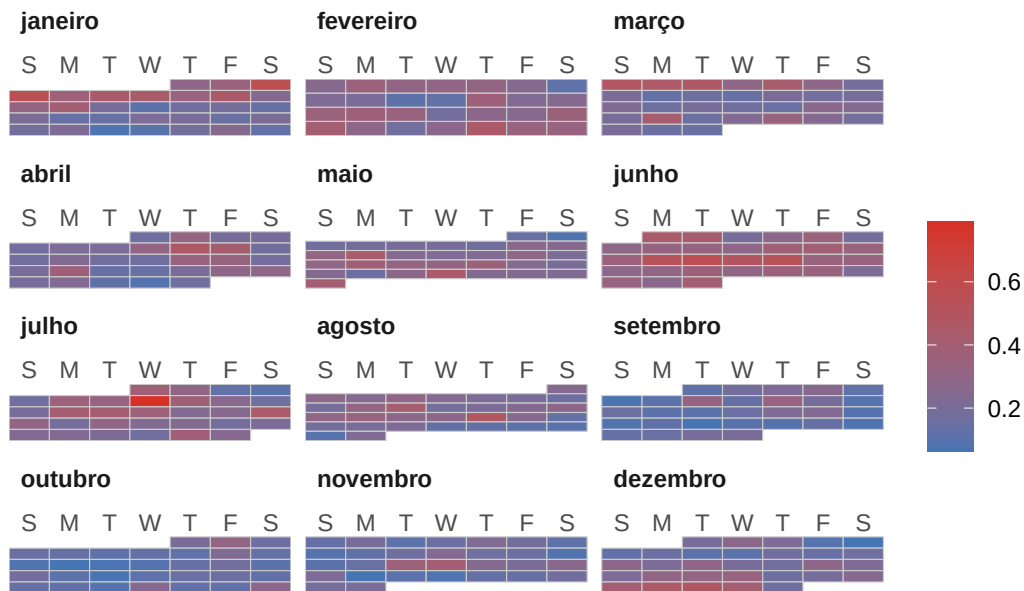
```
cDL + pal + ggtitle("Calendário de Atrasos – Delta Air Lines (DL)")
```

Calendário de Atrasos – Delta Air Lines (DL)



```
cUA + pal + ggtitle("Calendário de Atrasos - United Airlines (UA)")
```

Calendário de Atrasos – United Airlines (UA)



```
cUS + pal + ggtitle("Calendário de Atrasos - US Airways (US)")
```

Calendário de Atrasos – US Airways (US)

